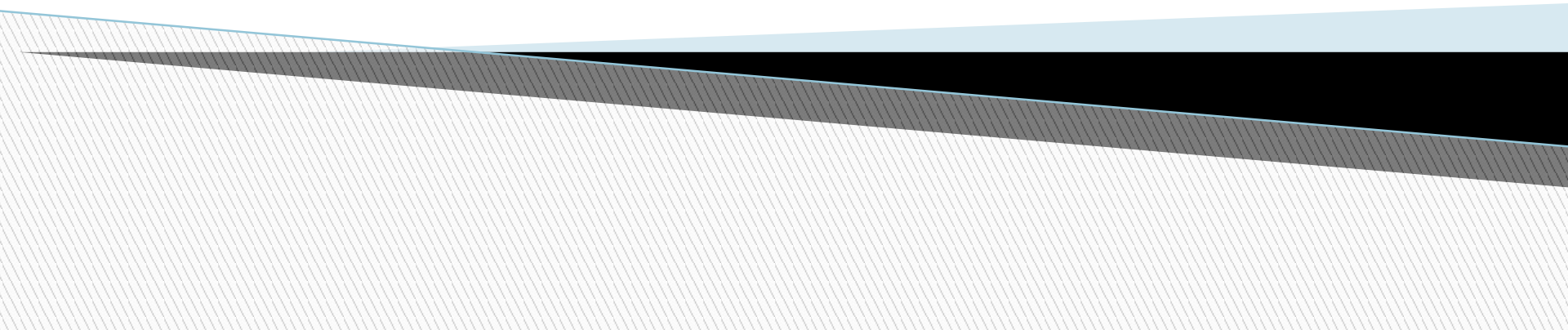


CLOSED ECONOMY

Week9-10



Outline

- Two-sectors economy
 - Consumption and saving
 - Investment
 - National income equilibrium and multiplier model
- Three-sectors economy
 - The role of fiscal policy in the multiplier model

Consumption

- ❑ Consumption is an activity that destroyed utility of goods and services.
- ❑ Personal consumption expenditure is the expenditure (*pengeluaran*) of households to purchase non durable and durable goods (except new housing) and services.
- ❑ A linear consumption function:

$$C = C_0 + bY$$

where, C_0 = autonomous consumption; b = marginal propensity to consume (MPC), and $0 < b < 1$; Y = disposable income

Saving

- Personal saving is the part of disposable income that is not consumed
- Therefore, the saving function can be derived as follows:

$$S = Y - C$$

$$S = Y - C_0 + bY$$

$$S = -C_0 + (1 - b)Y$$

where, $-C_0$ = dissaving (or negative saving);
 $(1-b)$ = marginal propensity to save (MPS),
and $0 < b < 1$

Income, Consumption and Saving

	Disposable Income	Consumption	Saving
	(1)	(2)	(3) = (1)–(2)
A	24,000	24,200	- 200
B	25,000	25,000	0
C	26,000	25,800	200
D	27,000	26,600	400
E	28,000	27,400	600
F	29,000	28,200	800
G	30,000	29,000	1,000

Income, Consumption and Saving

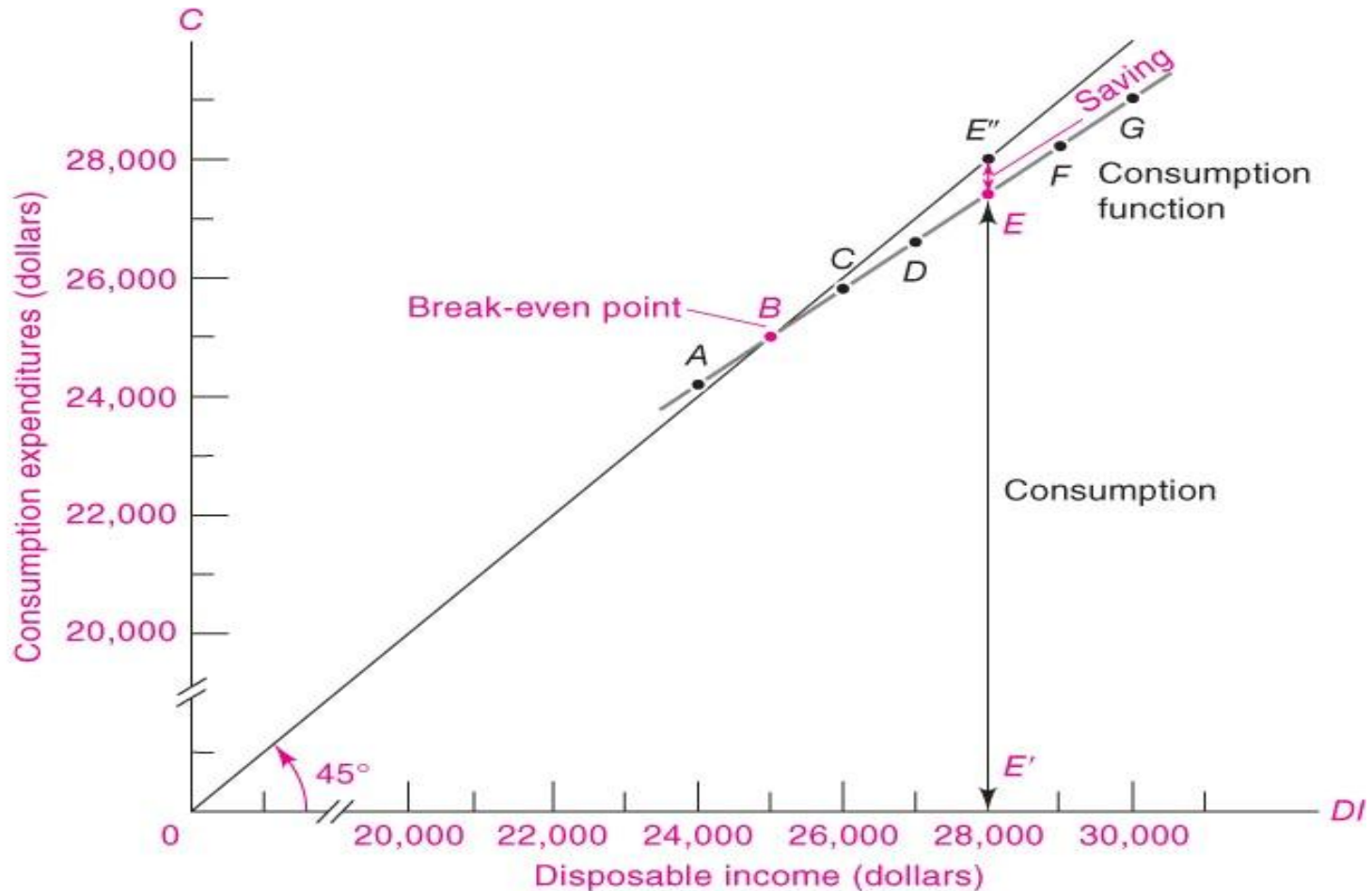


FIGURE 21-2. A Plot of the Consumption Function

Income, Consumption and Saving

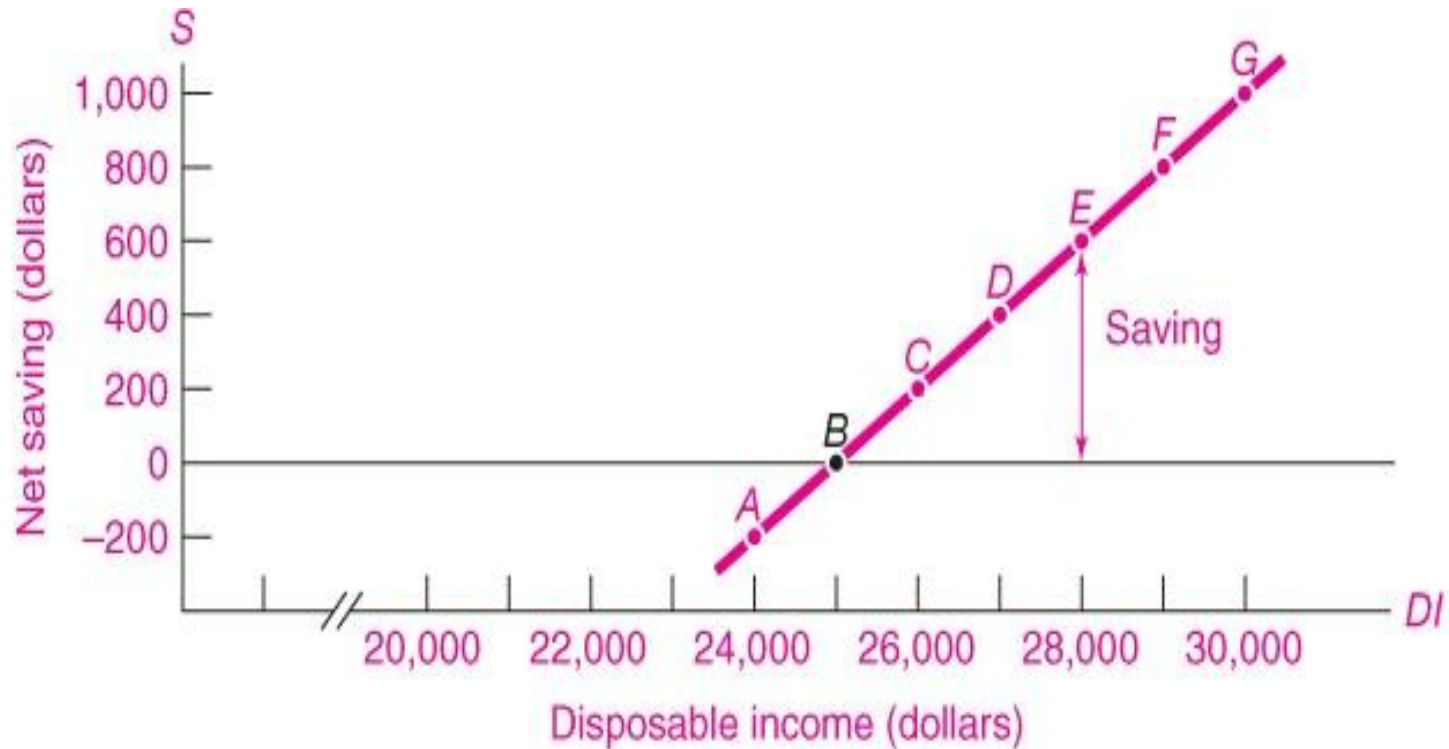


FIGURE 21-3. The Saving Function Is the Mirror Image of the Consumption Function

MPC and MPS

- MPC is the slope of the consumption function which measures the additional or extra consumption that results from an extra dollar of disposable income
- MPS is the slope of the saving function which measures the fraction of an additional or extra dollar of disposable income that goes to extra saving
- The relation between MPC and MPS
 - $MPC + MPS = 1$

MPC and MPS

	(1) Disposable income (after taxes) (\$)	(2) Consumption expenditure (\$)	(3) Marginal propensity to consume <i>MPC</i>	(4) Net saving (\$) (4) = (1) - (2)	(5) Marginal propensity to save <i>MPS</i>
A	24,000	24,200	$800/1,000 = 0.80$	-200	$200/1,000 = 0.20$
B	25,000	25,000		0	$200/1,000 = 0.20$
C	26,000	25,800	$800/1,000 = 0.80$	200	$200/1,000 = 0.20$
D	27,000	26,600	$800/1,000 = 0.80$	400	$200/1,000 = 0.20$
E	28,000	27,400	$800/1,000 = 0.80$	600	$200/1,000 = 0.20$
F	29,000	28,200	$800/1,000 = 0.80$	800	$200/1,000 = 0.20$
G	30,000	29,000	$800/1,000 = 0.80$	1,000	$200/1,000 = 0.20$

TABLE 21-4. The Marginal Propensities to Consume and to Save

MPC and MPS

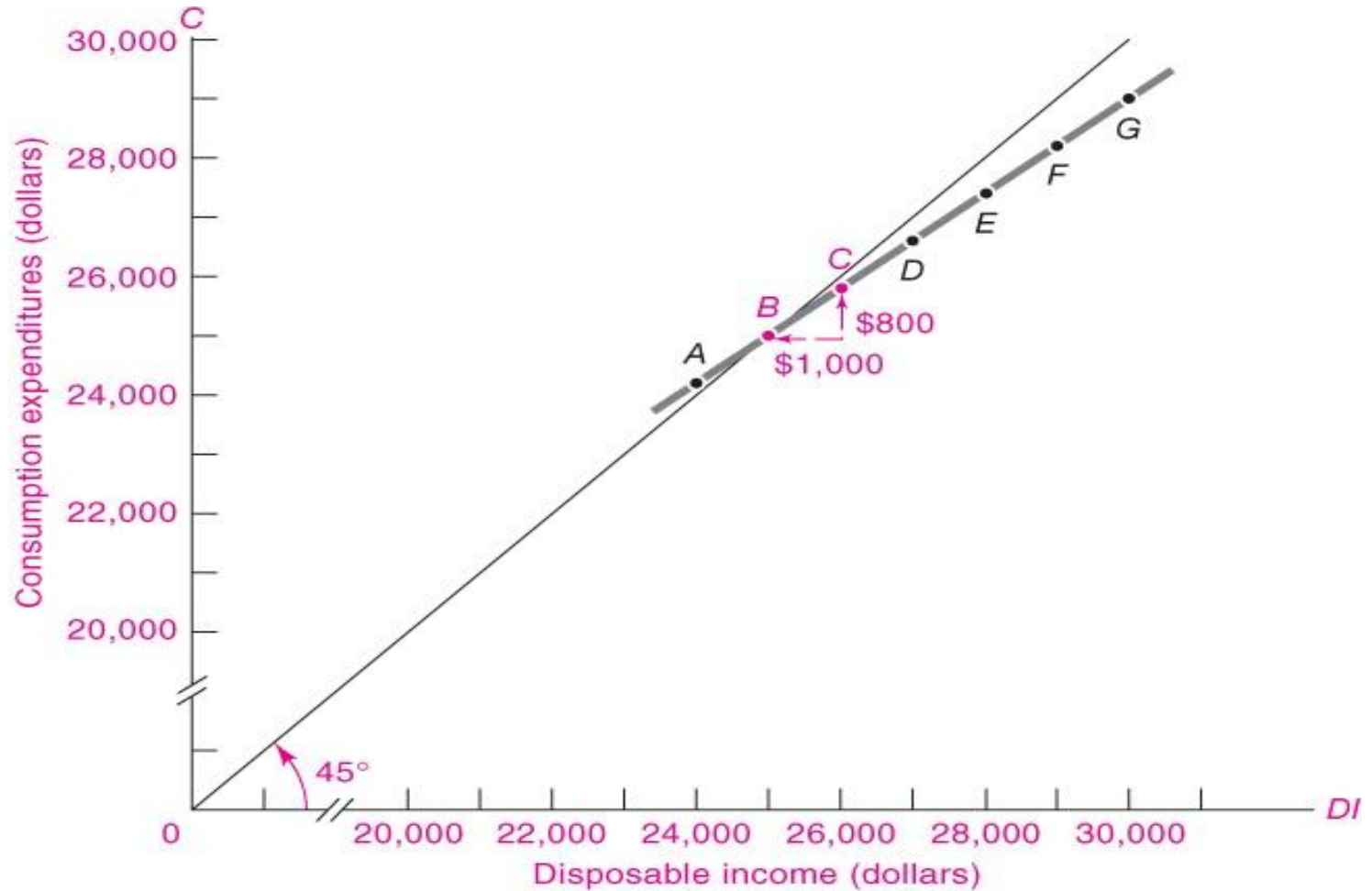


FIGURE 21-4. The Slope of the Consumption Function Is Its *MPC*

$C = \text{Autonomous Consumption} + \text{Induced Consumption}$

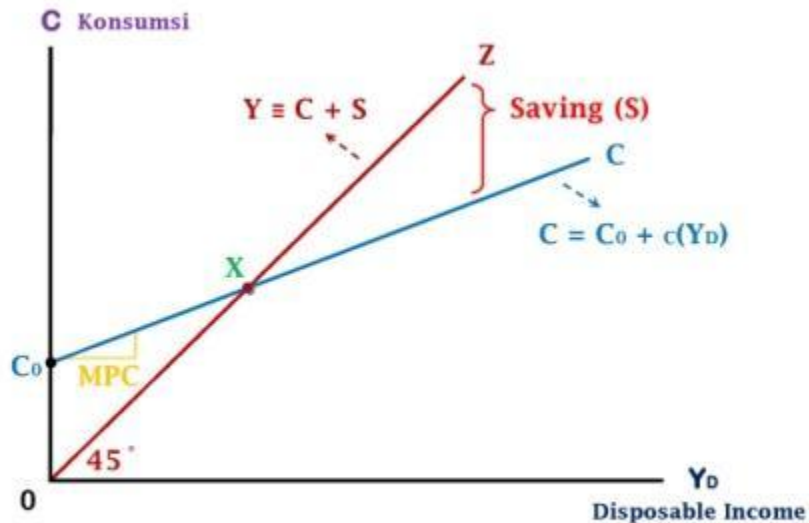
$$C = C_0 + cY_D$$

$$C = C_0 + c(Y - T)$$

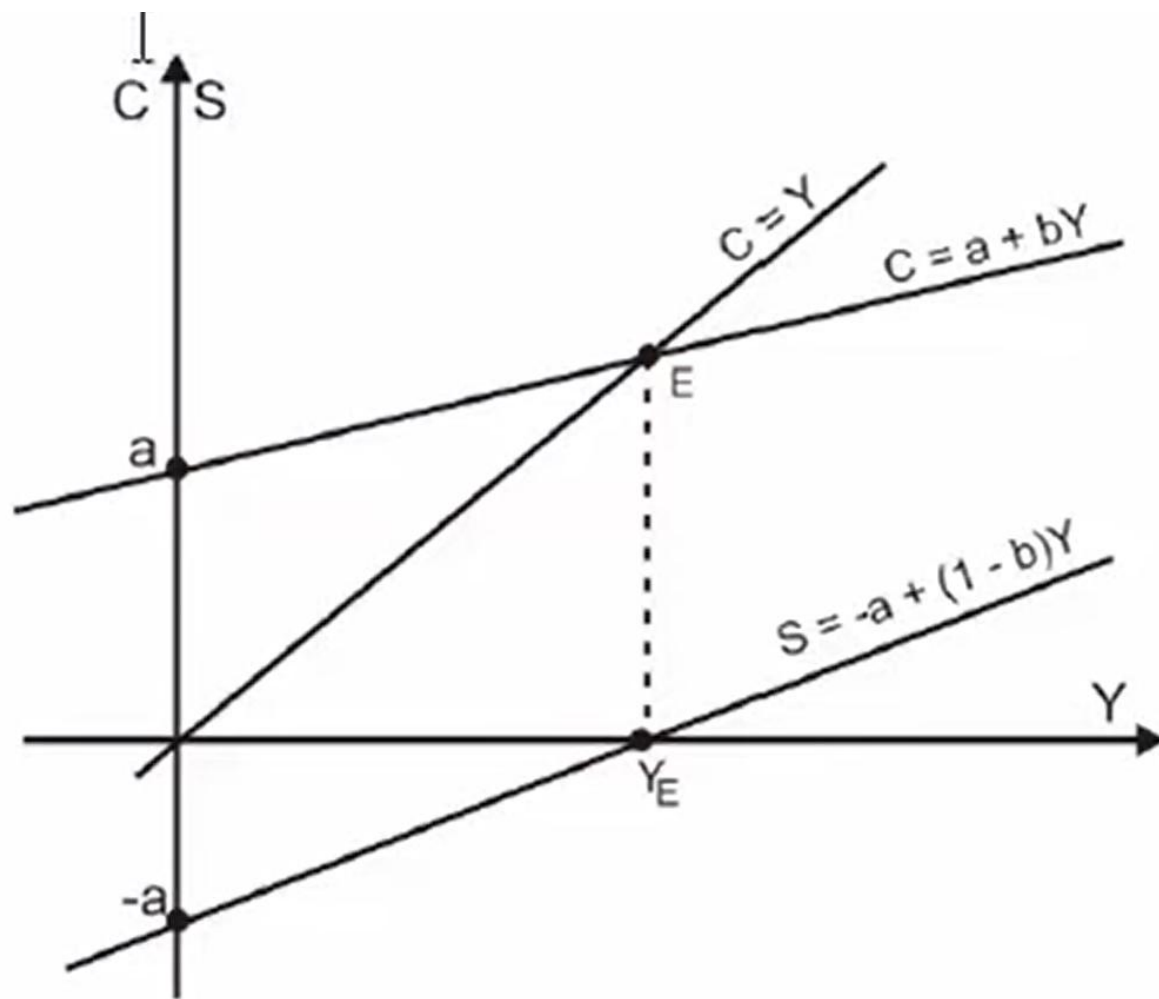
$$MPC = \frac{\Delta C}{\Delta Y}$$

$$MPC = \frac{C_2 - C_1}{Y_2 - Y_1}$$

$$MPS = \frac{\Delta S}{\Delta Y_d} = \frac{S_2 - S_1}{Y_2 - Y_1}$$



Gambar 1. Kurva Fungsi Konsumsi.



Investment

- Macroeconomist use the term ‘investment’ or ‘real investment’ to mean additions to the stock of productive assets or capital goods (=the purchase of new capital) such as equipment or buildings.
 - When Amazon.com builds a new warehouse or when the Smiths build a new house, these activities represent investment

Investment

- Many people speak of “investing” when buying a piece of land, stock, or using money to open a saving account.
 - In economics, these purchases are really financial transactions or “financial investments,” because what one person is buying, someone else is selling, and the net effect is zero.
 - There is investment only when real capital is produced

Istilah BPS

- ❑ Investasi mencakup pembentukan modal tetap bruto dan perubahan stok.
- ❑ Pembentukan modal tetap bruto (PMTB) = *Gross fixed capital formation* (GFCF) adalah pengeluaran untuk barang modal yang mempunyai umur pemakaian lebih dari satu tahun dan tidak merupakan barang konsumsi.
 - PMTB mencakup bangunan tempat tinggal dan bukan tempat tinggal, bangunan lain seperti jalan dan bandara, serta mesin dan peralatan.
 - Pengeluaran barang modal untuk keperluan militer tidak dicakup dalam rincian ini tetapi digolongkan sebagai konsumsi pemerintah.

Cont'd

❓ Perubahan stok

- Stok terdiri dari bermacam-macam barang yang akan dipakai, yang sedang dalam proses pengerjaan dan barang-barang yang sudah jadi tetapi belum dijual. Dengan demikian pemegang stok sebagian besar adalah perusahaan, termasuk perusahaan pemerintah dan pemerintah sendiri. Yang merupakan stok pemerintah adalah persediaan barang-barang strategis, seperti bahan pangan. Nilai perubahan stok pada tahun tertentu adalah selisih antara stok akhir tahun dengan stok awal tahun. Data perubahan stok barang-barang strategis pemerintah belum tersedia.

Important to note:

- Capital consist of durable produced items that are in turn used as productive inputs for further production.
 - **In an earlier era**, capital consisted primarily of tangible assets
 - Three important categories of tangible capital are **structures** (such as factories and home), **equipment** (such as automobile, machine tools and trucks), **inventories** (such as cars in dealers' lots)
 - **But today**, intangible capital or 'e-capital' (such as computer software, patents, and brand names) is increasingly important.

Important to note:

- We must distinguish tangible and intangible assets from financial assets.
 - Financial assets are essentially pieces of paper or electronic records.
 - More precisely, financial assets are monetary claims by one party against another party
 - An example is a mortgage, which is a claim against a homeowner for monthly payments of interest and principal; these payments will repay the original loan that helped finance the purchase of the house.
 - In this case, a tangible asset will serve as collateral for financial asset.

Important to note:

- It is clear that tangible assets are an essential part of an economy because they increase the productivity of other factors.
- **But what function do financial assets serve?**
 - These assets are crucial because of the mismatch between saver and investors

Important to note:

- The rate of return on investment is the annual net dollar return (revenue less expenses) per year for every dollar of invested capital.
 - An example: a rental car company 'KARTIKA' buys a used car for \$20,000 and rent it out. After subtracting all expenses (revenue less wages, office supplies, energy cost, etc) and assuming no change in the car's price, this company earns a net rental of \$2400 each year. Thus, the rate of return is
$$\$2,400/\$20,000 = 12\%$$
- Interest rate are the rate of return on financial assets, measured in percent per year.

Important to note:

- **The present value of assets** is the dollar value today of a stream of future income
- It is measured by calculating how much money invested today would be needed, at going interest rate, to generate the asset's future stream of receipts.
- General formula for present value

$$V = \frac{N_1}{1+i} + \frac{N_2}{(1+i)^2} + \dots + \frac{N_t}{(1+i)^t}$$

where i = interest rate, and N = net receipts

Important to note:

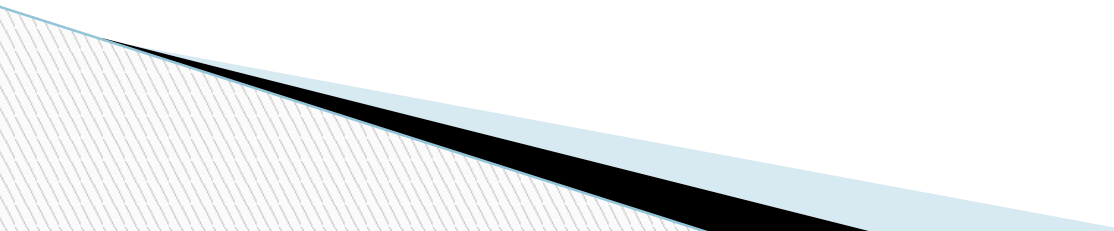
❓ Interest Rates and Asset Prices

- Asset prices tend to move inversely with interest rates because their present value decrease as the interest rate increases (see examples in Samuelson (ch.15 p.287))
- The dependence of asset prices on interest rate is a **general property of financial assets**.
- The price of stocks, bonds, real estate, and many other long-lived assets will decline as interest rates rise.
- Harga saham, obligasi, real estat, dan banyak aset berumur panjang lainnya akan turun seiring dengan kenaikan suku bunga.

❓ Real vs. Nominal Interest Rates

- ❓ Nominal interest rate (sometime also called the money interest rate) is the interest rate on money in terms of money
- ❓ Real interest rate is the nominal interest rate after corrected for inflation

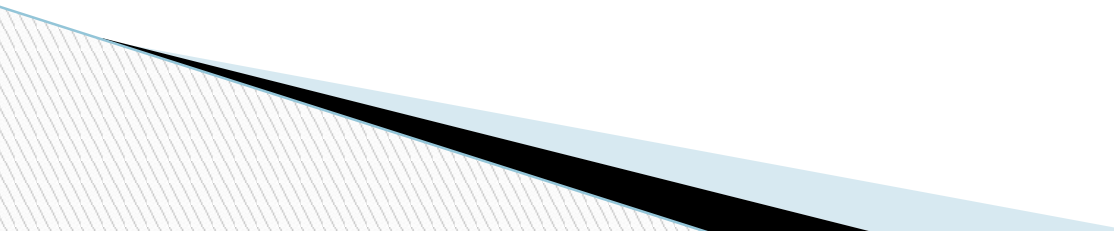
The roles of investment in macroeconomics

- ❑ Effecting short run output through its impact on aggregate demand.
 - ❑ Influencing long run output growth through the impact of capital accumulation.
 - ❑ Mempengaruhi output jangka pendek melalui dampaknya terhadap **permintaan agregat**.
 - ❑ Mempengaruhi pertumbuhan output jangka panjang melalui dampak **akumulasi modal**.
- 

Determinants of Investment

- ❑ Revenue: the overall level of output (or GDP)
- ❑ Cost: price of the capital good, interest rate, taxes
- ❑ Expectation: profit expectations and business confidence

pendapatan, kos, ekspektasi



Investment Demand Curve

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Project	Total investment in project (\$, million)	Annual revenues per \$1,000 invested (\$)	Cost per \$1,000 Borrowed at Annual Interest Rate of:		Annual Net Profit per \$1,000 Borrowed at Annual Interest Rate of:	
					10%	5%
			10%	5%		
			(\$)	(\$)	(\$)	(\$)
					(6) = (3) – (4)	(7) = (3) – (5)
A	1	1,500	100	50	1,400	1,450
B	4	220	100	50	120	170
C	10	160	100	50	60	110
D	10	130	100	50	30	80
E	5	110	100	50	10	60
F	15	90	100	50	–10	40
G	10	60	100	50	–40	10
H	20	40	100	50	–60	–10

TABLE 21-5. The Profitability of Investment Depends on the Interest Rate

Investment Demand Curve

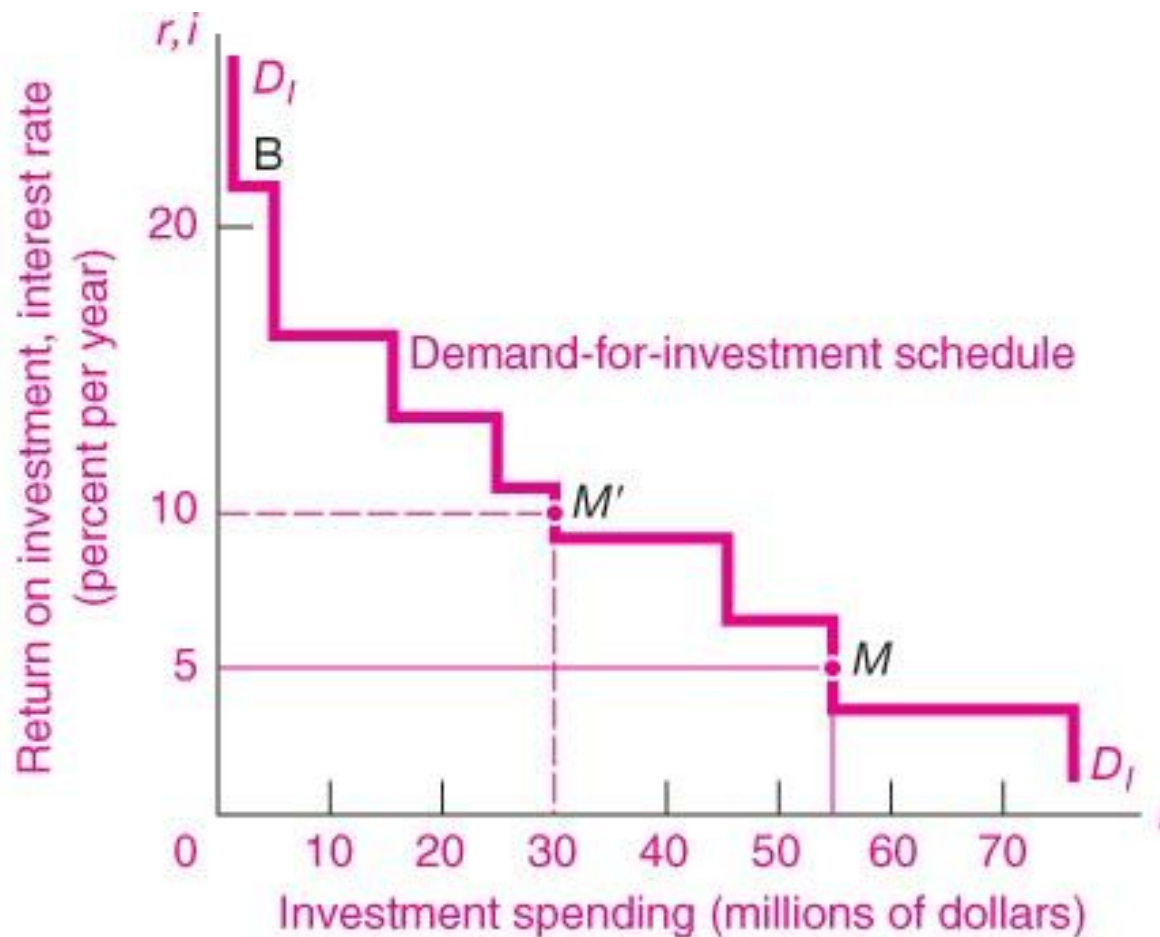


FIGURE 21-8. Investment Depends upon Interest Rate

Investment Demand Curve

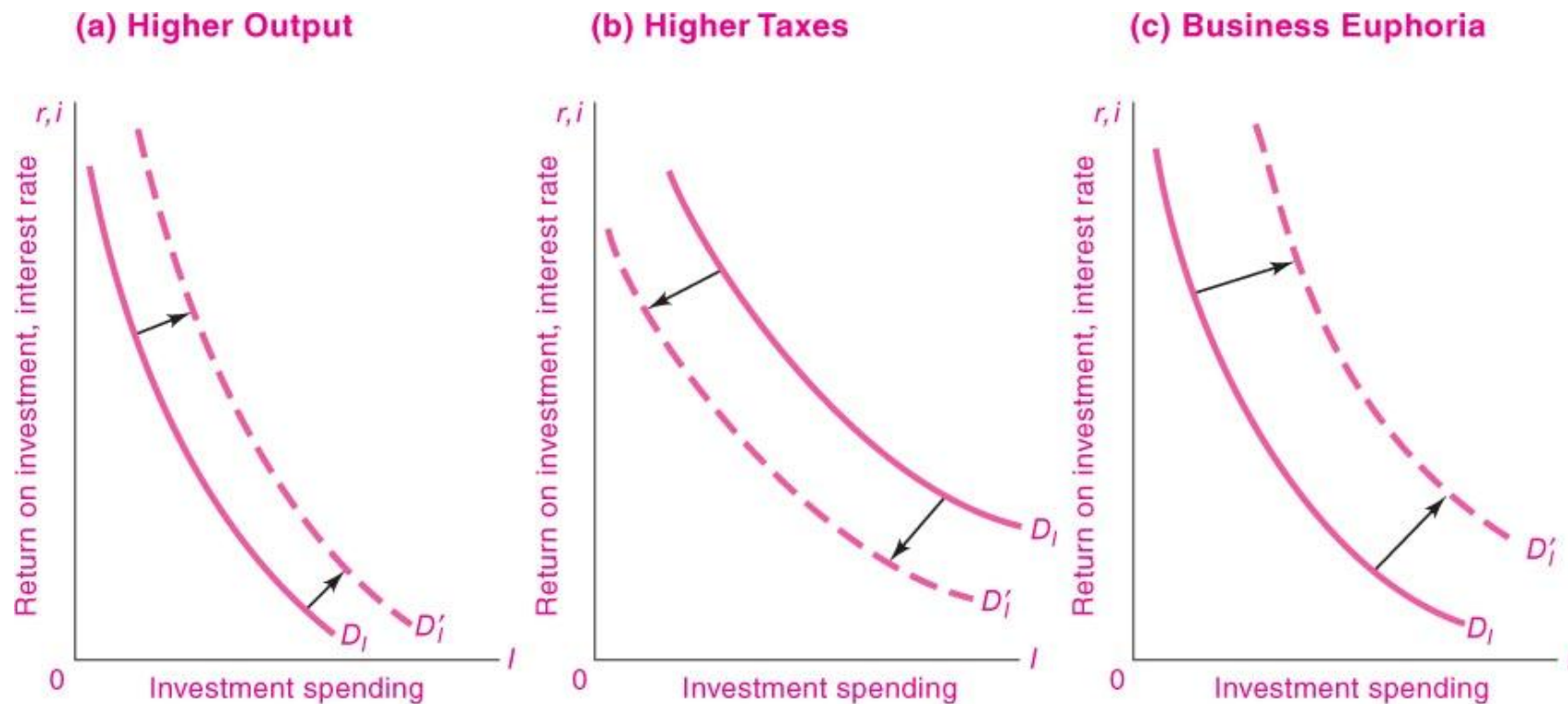


FIGURE 21-9. Shifts in Investment Demand Function

Equilibrium Output in the two-sectors economy

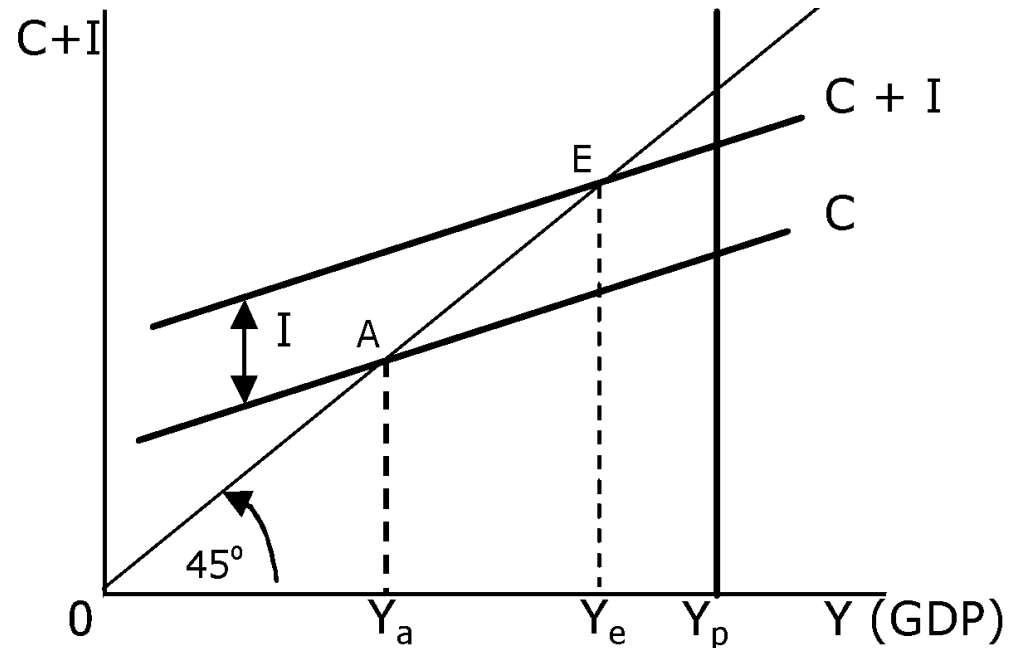
1. $Y = C + I$ approach

Mathematically:

$$Y = C + I$$

$$Y = C_0 + bY + I$$

$$Y = 1/(1-b) (C_0 + I)$$



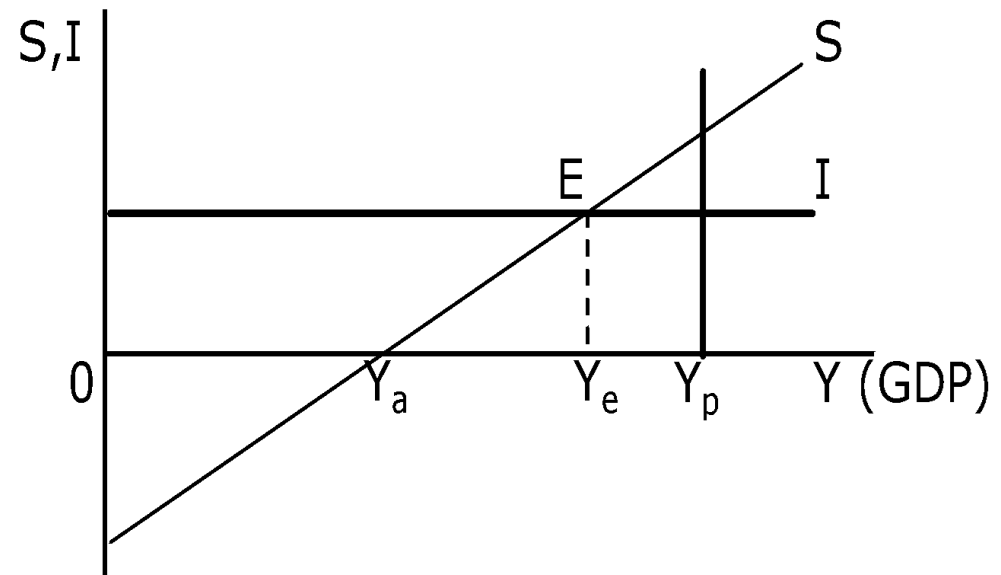
Equilibrium Output in the two-sector economy

2. $S = I$ approach Mathematically

$$I = S$$

$$I = -C_0 + (1 - b)Y$$

$$Y = 1/(1-b) (C_0 + I)$$



Equilibrium Output in the two-sector economy

GDP Determination Where Output Equals Planned Spending
(billions of dollars)

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Levels of GDP and <i>DI</i>	Planned consumption	Planned saving (3) = (1) - (2)	Planned investment	Level of GDP (5) = (1)	Total planned consumption and investment, <i>TE</i> (6) = (2) + (4)	Resulting tendency of output
4,200	3,800	400	200	4,200	> 4,000	Contraction
3,900	3,600	300	200	3,900	> 3,800	Contraction
3,600	3,400	200	200	3,600	= 3,600	Equilibrium
3,300	3,200	100	200	3,300	< 3,400	Expansion
3,000	3,000	0	200	3,000	< 3,200	Expansion
2,700	2,800	-100	200	2,700	< 3,000	Expansion

TABLE 22-2. Equilibrium Output Can Be Found Arithmetically at the Level Where Planned Spending Equals GDP

The Multiplier Model

- The multiplier model explain that each 'dollar' change in exogenous expenditure leads to a multiplied change in GDP.
- Key assumption:
 - The wages and prices are fixed
 - There are unemployed resources

Investment Multiplier

Suppose that MPC is $\frac{2}{3}$. How much GDP will change if investment in the economy increases by 1,000 billion rupiahs?

Investment Multiplier (arithmetic approach)

1000	=	1	X	1000
+			+	
666,67	=	$2/3$	X	1000
+			+	
444,44	=	$(2/3)^2$	X	1000
+			+	
296,30	=	$(2/3)^3$	X	1000
+			+	
197,53	=	$(2/3)^4$	X	1000
+			+	
.			.	
.			.	
.			.	
.			.	
3000	=	$1/(1 - 2/3)$	X	1000

Investment Multiplier (arithmetic approach)

Change in GDP

$$= (1 + 2/3 + 2/3^2 + 2/3^3 + 2/3^4 + 2/3^n) 1000$$

$$= 1/(1 - 2/3) \times 1000 = 3000$$

The simple multiplier formula is

$$\text{Change in GDP} = 1/(1 - \text{MPC}) \times \text{change in investment}$$

or

$$= 1/\text{MPS} \times \text{change in investment}$$

Investment Multiplier

$$Y = C + I$$

$$Y = C_0 + bY + I$$

$$Y = 1/(1-b) (C_0 + I)$$

$$Y + \Delta Y = 1/(1-b) (C_0 + I + \Delta I)$$

$$\Delta Y = 1/(1-b) \Delta I$$

$$I = S$$

$$I = -C_0 + (1-b)Y$$

$$I + \Delta I = -C_0 + (1-b)(Y + \Delta Y)$$

$$I + \Delta I = -C_0 + (1-b)Y + (1-b)\Delta Y$$

$$\Delta Y = 1/(1-b) \Delta I$$

Investment Multiplier
Coefficient



Fiscal Policy in the Multiplier Model

- The role of fiscal policy in the economy
 - Allocative
 - Distributive
 - Stabilizer
- Instruments of fiscal policy
 - Government spending
 - Taxation

Important to note:

(SAM, ch. 16 p. 312-320)

❓ Principles of taxation

- Benefit vs. Ability to pay principles
 - The **benefit principle of taxation** taxes people in proportion to the marginal benefits they receive from a governmentally provided good.
 - The **ability-to-pay principle** suggests that the fairest tax is one based on your financial ability to support government activities.
- Horizontal and vertical equity
 - **Vertical equity** asserts that people better able to pay higher taxes should do so.
 - **Horizontal equity** requires that individuals who are equal in all important respects be treated equally.

Important to note:

- Pragmatic compromise in taxation
 - Progressive □ higher incomes are taxed proportionally more than lower incomes
 - regressive □ the percentage of income paid as taxes falls as income rises.
 - proportional taxes □ a fixed percentage of income

□ Classification of tax

- Indirect taxes
- Direct taxes

Equilibrium Output in the three-sectors economy

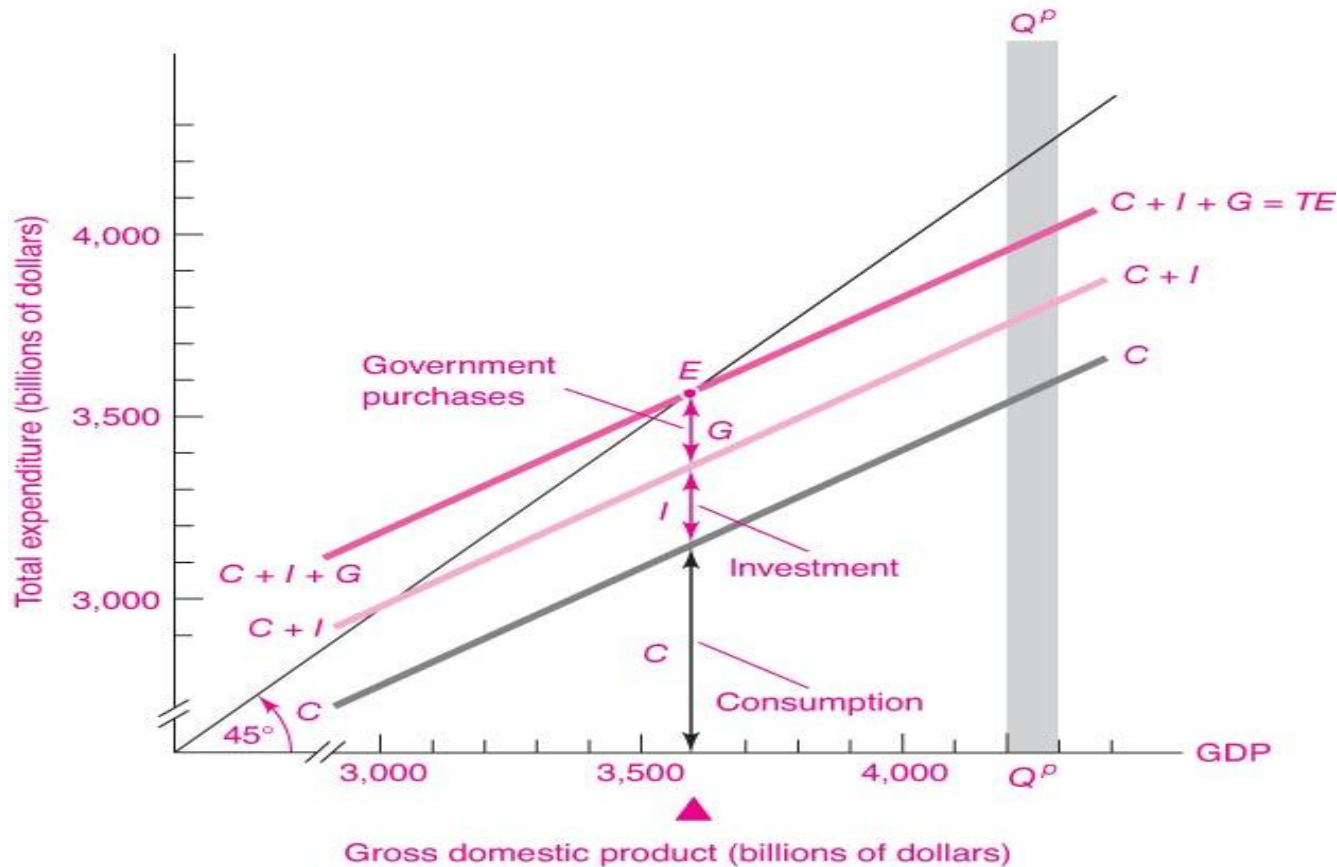


FIGURE 22-10. Government Purchases Add On Just like Investment to Determine Equilibrium GDP

Equilibrium Output in the three-sectors economy

Output Determination with Government Spending
(billions of dollars)

(1) Initial level of GDP	(2) Taxes T	(3) Disposable income DI	(4) Planned consumption C	(5) Planned investment I	(6) Government expenditure G	(7) Total planned expenditure, TE ($C + I + G$)	(8) Resulting tendency of economy
4,200	300	3,900	3,600	200	200	4,000	↓
3,900	300	3,600	3,400	200	200	3,800	↓
3,600	300	3,300	3,200	200	200	3,600	Equilibrium
3,300	300	3,000	3,000	200	200	3,400	↑
3,000	300	2,700	2,800	200	200	3,200	↑

TABLE 22-3. Government Purchases, Taxes, and Investment Also Determine Equilibrium GDP

Equilibrium Output

Assumption: tax is independence to income ($T = T_0$)

■ $Y = C + I + G$

$$Y = C_0 + bY_d + I + G$$

$$Y = C_0 + b(Y - T_0) + I + G$$

$$Y = C_0 + bY - bT_0 + I + G$$

$$Y = 1/(1-b) (C_0 - bT_0 + I + G)$$

$$Y = \frac{1}{1-b} (C_0 - bT_0 + I + G)$$

■ $I + G = S + T$

$$I + G = -C_0 + (1-b)Y_d + T_0$$

$$I + G = -C_0 + (1-b)(Y - T_0) + T_0$$

$$I + G = -C_0 + (1-b)Y + bT_0$$

$$Y = 1/(1-b) (C_0 - bT_0 + I + G)$$

$$Y = C + S + T$$

$$Y_d = Y - T$$

Fiscal Policy Multiplier

Assumption: tax is independence to income ($T = T_0$)

Government expenditure multiplier

$$Y = 1/(1-b) (C_0 - bT_0 + I + G)$$

$$Y + \Delta Y = 1/(1-b) (C_0 - bT_0 + I + G + \Delta G)$$

$$\Delta Y = 1/(1-b) \Delta G$$

where: $1/(1-b)$ is government expenditure multiplier

Tax multiplier

$$Y = 1/(1-b) (C_0 - bT_0 + I + G)$$

$$Y + \Delta Y = 1/(1-b) (C_0 - bT_0 - b\Delta T_0 + I + G)$$

$$\Delta Y = -b/(1-b) \Delta T_0$$

where: $-b/(1-b)$ is tax multiplier.

Example

$$\text{Multiplier } G = \frac{1}{1 - MPC(1 - t)}$$

□ Suppose that:

- $C = 300 + 0.75Y_d$
- $I = 400$
- $G = T = 200$
 - Then, GDP equilibrium will be
 $Y = (1/0.25)(300 - 150 + 400 + 200) = 3,000$
 - Government expenditure multiplier is 4. Therefore, an Increase in G by 50 will increase Y by 200
 - Tax multiplier is 3 (negative in value), so an increase in tax by 50 will decrease Y by 150

▶ $Y = C + I + G$

▶ $C = 300 + 0,75(Y - T)$

▶ $T = 200 + 0,15Y$

▶ $C = 300 + 0,75(Y - 200 - 0,15Y)$

▶ $Y = 1 / (0,25 + 0,1125(300 - 150 + 400 + 200))$

▶ $= 2,068,96$

▶ $G = 1 / 1 - MPC (1 - t)$

▶ $= 1 / 0,25 + 0,1125 = 2,758$

Equilibrium Output

Assumption: tax is dependence to income ($T = T_0 + tY$)

❓ $Y = C + I + G$

$$Y = C_0 + bY_d + I + G$$

$$Y = C_0 + b(Y - T_0 - tY) + I + G$$

$$Y = C_0 + bY - bT_0 - btY + I + G$$

$$Y = 1/(1-b+bt) (C_0 - bT_0 + I + G)$$

❓ $I + G = S + T$

$$I + G = -C_0 + (1 - b)Y_d + (T_0 + tY)$$

$$I + G = -C_0 + (1 - b)(Y - T_0 - tY) + (T_0 + tY)$$

$$I + G = -C_0 + (1 - b)Y - (1 - b)T_0 - (1 - b)tY + (T_0 + tY)$$

$$I + G = -C_0 + (1 - b)Y + bT_0 + btY$$

$$Y = 1/(1-b+bt) (C_0 - bT_0 + I + G)$$

Fiscal Policy Multiplier

Assumption: tax is dependence to income ($T = T_0 + tY$)

❓ Government expenditure multiplier

- $\Delta Y / \Delta G = 1 / (1 - b + bt)$

❓ Tax multiplier

- $\Delta Y / \Delta T_0 = -b / (1 - b + bt)$

- Impact of the change in tax rate (t) on the change in GDP

- $\Delta Y / \Delta t = -b / (1 - b + bt) Y$

Example

❓ Suppose that:

- $C = 300 + 0.75Y_d$
- $I = 400$
- $G = 200$
- $T = 200 + 0.15Y$

- Then, GDP equilibrium will be

$$Y = [1/(0.25+0.1125) (300 - 150 + 400 + 200)] = 2,068.96$$

- Government expenditure multiplier is 2.758. Thus, an Increase in G by 50 will increase Y by 137.93
- Tax multiplier is 2.069 (negative in value), so an increase in tax by 50 will decrease Y by 103.45

Important to note:

- National income identity in the closed economy

$$Y = C + I + G$$

- Subtraction C and G from both side of the equation, we obtain:

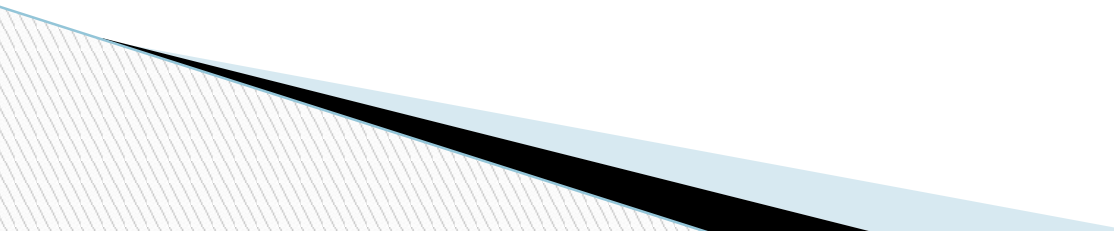
$$Y - C - G = I$$

- Manipulating this equation to obtain

$$(Y - T - C) + (T - G) = I$$

Where the left hand side is national saving, which consists of **private saving** ($Y - T - C$) and **public saving** ($T - G$)

Important to note:

- The last equation reveals an important fact:
 - For the economy as a whole, saving must be equal to investment. (the mechanisms lie behind this identity will be discussed in the next topic)
 - The larger the consumption, the smaller private saving, and the result would be lower national saving
 - when the government spends more than it receives in tax revenue, the resulting budget deficit lower national saving
- 

Example

- Suppose GDP is Rp 8.00 trillion, taxes are Rp 1.50 trillion, private saving is Rp 0.50 trillion, and public saving is Rp 0.20 trillion. Calculate consumption, government purchases, national saving, and investment.

- **Consumption (C)**

- Private saving = $Y - T - C$

- $0.50 \text{ trillion} = 8.00 \text{ trillion} - 1.50 \text{ trillion} - C$

- $C = 6.00 \text{ trillion}$

Example

- **Government purchases (G)**
 - Public saving = $T - G$
 $0.20 \text{ trillion} = 1.50 \text{ trillion} - G$
 $G = 1.30 \text{ trillion}$
- **National saving** = private saving + public saving
= $0.50 \text{ trillion} + 0.20 \text{ trillion}$
= 0.70 trillion
- Equilibrium condition require that national saving equal to investment. Thus **investment** must be 0.70 trillion .
 - Alternatively, we can use national income identity (recall: $Y = C + I + G$) to obtain investment by subtracting GDP (Y) with C and G.

Thank you
for your attention

